# Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

## Exercise 1.1 · Getting to grips with indices

Consider the sequence $(u_n)$ defined for every natural number $n$ by $u_n = 2n^2 - 3n$. The first term is therefore the one with index 0.

1. Calculate separately $u_0$, $u_1$ and $u_4$, by showing the substitution of $n$.
2. Express $u_{n+1}$ as a function of $n$, then expand and simplify this expression.
3. Calculate $u_{n+1} - u_n$. By studying its sign for integer indices, describe the variation of the sequence.

## Exercise 1.2 · Choosing the right model

A nursery currently has 100 young plants. It is considering two models to forecast its numbers: in model A, the number increases by 15 plants per month; in model B, it increases by 10 per cent per month. Let $a_n$ and $b_n$ the projected numbers after $n$ months, with $a_0 = b_0 = 100$. The two models are studied separately.

1. **Model A.** Write the relationship between $a_{n+1}$ and $a_n$. Identify the type of sequence, then express $a_n$ as a function of $n$. Calculate $a_2$.
2. **Model B, after the study of geometric sequences.** Answer the same questions for $(b_n)$ and calculate $b_2$.
3. For each model, explain what is assumed constant. Give one reason why a very long-term forecast could become unrealistic.

## Exercise 1.3 · Recovering a sequence

The sequence $(u_n)$ is geometric, defined from index 0. Its common ratio $q$ is strictly positive. We know only $u_1 = 6$ and $u_3 = 54$.

1. Determine $q$ by linking the two known terms. Show all the steps that make it possible to choose its value.
2. Deduce $u_0$, then $u_5$.
3. What would change if the hypothesis $q > 0$ were removed? Give all possible ratios and explain whether the value of $u_5$ would change.

## Exercise 1.4 · Rows of seats

A concert hall has 12 rows. The first row has 18 seats; each following row has two more seats than the previous one. Let $u_n$ be the number of seats in row $n$, for $1 \leq n \leq 12$.

1. Define the sequence $(u_n)$ and calculate the number of seats in the last row.

2. Calculate the total capacity of the hall. Explain the number of terms used in the sum.

3. Calculate the mean number of seats per row and use this result to check the total found.

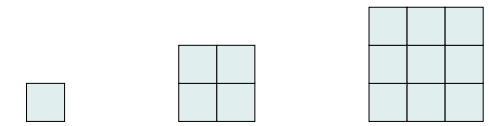
## Exercise 1.5 · Adding up squares

Patterns are built with identical square tiles. Pattern number 1 contains one tile. To go from pattern number $n$ to pattern number $n+1$, we add $2n+1$ tiles. Let $u_n$ be number of tiles in pattern number $n$, for $n \geq 1$.

1. Calculate $u_2$ and $u_3$. Draw an arrangement of these tiles that suggests a formula for $u_n$.

2. Conjecture an expression for $u_n$ as a function of $n$.

3. Check that this expression gives the first term and satisfies the construction rule. Explain why these two checks identify the sequence.



Stage 1 Stage 2 Stage 3

A possible layout for the first three patterns; the drawing helps to formulate a conjecture, which remains to be justified.

## Exercise 1.6 · Two repayment plans

A person must repay 600 monetary units, interest-free. Two plans are being considered. With plan A, she pays 50 units at the end of each month until the loan is repaid. With plan B, she pays each month half of what she still owes. The calculations for model B are done without rounding. Let $a_n$ and $b_n$ be the debts remaining after the payment for month $n$, with $a_0 = b_0 = 600$.

1. Express $a_n$ until the debt is paid off. In which month is the repayment completed?

2. Express $b_n$. Determine the first month in which the remaining debt is strictly less than 10 units ; check the previous month.

3. In model B without rounding, does the debt become exactly zero after a finite number of months? Justify and comment on the difference from a real situation.

## Exercise 1.7 · Reading a list

A program must store the terms of the sequence defined by $u_0 = 2$ and $u_{n+1} = 3u_n - 1$. It starts with the instructions `u=2` and `L=[u]`. It then executes, five times and in this order, `u=3*u-1` then `L.append(u)`.

1. Run the first two iterations of the loop by hand. Then give the contents of the list.
2. How many elements does the list contain after the five iterations? What is the index of the last term recorded?
3. Write the complete program in Python, then check the value of the last element. Explain why the number of terms is one more than the number of iterations.

### Exercise 1.8 · Delayed review

This question is done one week after the chapter, without looking at the formulas. A sequence is defined for $n \geq 2$ by $u_2 = 12$ and $u_{n+1} = u_n - 3$.

1. Calculate $u_7$ stating how many steps separate indices 2 and 7.
2. Calculate the sum $u_2 + u_3 + \cdots + u_7$. Explain the difference between the number of terms of this sum and the number of steps calculated in the first question.

**Independent work.** Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

## To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

**CQFD, exercise 9 p. 64, questions a and b.** Calculate the first four terms. Write the substitutions before simplifying.

**CQFD, exercise 11 p. 64, all questions.** Calculate the terms successively and keep a record of the four steps.